Part C — AI Usage Report

DSC 190/291 — Assignment 5 Student: Zeyu Bian

1. Where I used AI

- **A.2 reduction design.** I sketched the "one coordinate per set, M repeated negatives per universe element" idea myself from the hint, then iterated with AI on bookkeeping: which examples should be positive, what threshold K to set, and how $M = q + 1$ enters the contradiction in the reverse direction. AI was useful as a "type-checker" — it caught a sign error in my first draft (I had positive examples rewarding selection rather than charging it).
- **A.3 experiment.** I drafted the synthetic-data and brute-force-enumeration scaffolding myself. AI helped me decide what to plot (runtime vs. d and k, agreement vs. noise) and the right way to make the brute-force "approximate exactness" disclaimer precise (the per-support fit is convex, the enumeration is exact).
- **B.2 piecewise linear analysis.** I computed the three regions ($w \le -1/M$, $-1/M \le w \le 1$, $w \ge 1$) by hand. AI helped me state cleanly why $M > (1-p)/p$ is exactly the condition that makes the middle slope positive (so w^* is $-1/M$ and not somewhere in the interior).
- **B.3 rank argument.** I knew the orthogonality identity for characters but asked AI to double-check the matrix factorization $H = W\Phi$ direction (the rank inequality $\mathrm{rank}(W\Phi) \le \min(\mathrm{rank}(W), \mathrm{rank}(\Phi))$).
- **B.4 Jensen example.** I picked the $(1, 1), (-1, -1)$ pair myself. AI suggested the alternative $(2, 1), (1, 2)$ with midpoint $(1.5, 1.5)$, which doesn't violate Jensen here (vu is then 2, 2, 2.25); I rejected that suggestion and kept the symmetric sign-flip pair, where the midpoint is the zero predictor and the violation is sharp.

2. AI suggestions I accepted (and why)

- **"$M = q + 1$ exceeds the mistake budget" framing for A.2 reverse direction.** AI suggested stating the contradiction as "any uncovered u_i forces $M > k$ mistakes; the total mistake budget is k; therefore no uncovered u_i." I accepted because it makes the threshold $K = (q - k) + Mr$ feel inevitable rather than arbitrary.
- **Explicit sign-of-w_j accounting in A.2 forward direction.** I had originally only said "set $w_j = -1$ for $j \in R$" without explaining why this gives zero coverage mistakes. AI suggested writing out $\langle w, a^{(i)} \rangle = -|\{j \in R : u_i \in A_j\}| < 0$ explicitly. I accepted; the inequality is the load-bearing step.
- **Three-region case split for B.2 hinge minimization.** AI suggested partitioning $\mathbb{R}$ at the two hinge breakpoints rather than trying to optimize the sum directly via subgradients. I accepted because piecewise-linear minimization is cleaner for a one-dimensional problem.
- **Explicit $H = W\Phi$ factorization for B.3.** AI proposed writing the assumption $\chi_I(x) = \langle w_I, \varphi(x) \rangle$ as a $2^d \times 2^d$ matrix factorization through the D-dimensional space. I accepted because it makes the rank inequality $2^d = \mathrm{rank}(H) \le D$ a one-line corollary.

3. AI suggestions I rejected or substantially modified

- **A.2 "intercept variable" first draft.** AI's initial reduction introduced an intercept coordinate $x_0 = 1$ on every example, claiming this was needed to handle the tie-breaking convention. This was wrong: the homogeneous halfspace family $\mathcal{H}_{d,k}$ has no intercept by definition, and adding one *would* change the class. I removed the intercept and verified directly that the $\text{sign}(0) = +1$ convention handles the $w_j = 0$ case correctly: positive example $(e_j, +1)$ with $w_j = 0$ gives $\langle w, e_j \rangle = 0$, sign $= +1$, correctly classified.

- **A.3 "use Gurobi for an exact 0-1 baseline" suggestion.** AI proposed solving the exact mixed-integer 0-1 ERM with a commercial MIP solver. I rejected this because
  (a) it requires an extra dependency the grader may not have, and (b) the exponential enumeration of supports already illustrates the computational hardness — adding a MIP would not change the qualitative finding. I stated explicitly that the baseline is approximate-per-support, exact-over-supports.
- **B.2 "look at the dual" approach.** AI offered to derive the unique minimizer via the hinge-loss Lagrangian. I rejected; the three-region piecewise-linear analysis is direct and avoids machinery the reader hasn't seen for this problem.
- **B.3 Hadamard-matrix invocation.** AI's first proof wanted to identify H with the Hadamard matrix H_{2^d} and quote $H_{2^d} H_{2^d}^\top = 2^d I$. I modified this to a self-contained proof using $\chi_I \chi_J = \chi_{I \triangle J}$ and $\sum_x \chi_K(x) = 0$ for $K \neq \emptyset$, because the Hadamard identification (with the right row/column ordering) is itself a lemma the assignment doesn't grant.
- **B.4 Jensen with $(2, 1), (1, 2)$.** As noted in §1: AI's first attempt gave a *non*-violating example. I caught it by computing the three loss values; replaced with $(1, 1), (-1, -1)$ where the midpoint is $(0, 0)$ and Jensen fails by 1.

4. How I verified correctness

- **A.2 reduction (both directions).** Wrote a small Python check (not included in the submission since it duplicates the written proof): for several random SETCOVER instances with $q \leq 6$, $r \leq 6$, $k \leq 3$, constructed the AGREEMENT sample and verified that brute-force search over $\{-1, 0, +1\}^q$-valued weight vectors achieves $\geq K$ iff the original instance is a YES-instance.
- **A.3 experiment.** Verified `empirical_agreement` directly against the definition on a $d = 2$, $n = 4$ toy example. Verified the runtime scaling matches $\binom{d}{k}$ growth by computing the ratio $T(d = 20, k = 3)/T(d = 20, k = 2) \approx 6.3$ vs. theoretical $\binom{20}{3}/\binom{20}{2} = 1140/190 = 6.0$, which matches.
- **B.1 LP.** Verified the LP optimum equals the hinge-loss optimum by sampling: ran the LP via `scipy.optimize.linprog` on small random instances and checked the hinge-loss recomputed from $w = w^+ - w^-$ matches the LP optimum value (a side check, not in the submission).
- **B.2 piecewise minimum.** Verified the three slopes by direct algebra: $-(1 - p) < 0$, $pM - (1 - p) > 0$ (using $M > (1 - p)/p$), $pM > 0$. Checked $w^* = -1/M$ is the unique global minimizer because L decreases then increases with no flat regions (all three slopes are strictly nonzero).
- **B.3 rank.** Verified $\chi_I \chi_J = \chi_{I \triangle J}$ from the definition: $\chi_I(x)\chi_J(x) = \prod_{i \in I} x_i \prod_{j \in J} x_j = \prod_{i \in I \triangle J} x_i$, using $x_i^2 = 1$ to cancel each $i \in I \cap J$. Verified $\sum_{x \in \{\pm 1\}^d} \chi_K(x) = \prod_{i \in K} \sum_{x_i \in \{\pm 1\}} x_i = 0$ for $K \neq \emptyset$.
- **B.4 Jensen.** Computed the three loss values directly: $L_{\text{net}}(1, 1) = (1 \cdot 1 - 1)^2 = 0$, $L_{\text{net}}(-1, -1) = ((-1)(-1) - 1)^2 = 0$, $L_{\text{net}}(0, 0) = (0 - 1)^2 = 1$. $1 > \frac{1}{2} \cdot 0 + \frac{1}{2} \cdot 0$, confirmed.

AI workflow updates

1. **"Type-check reduction outputs" prompt.** When designing a reduction, I now ask AI to write out the full mistake/cost accounting for *both* directions before declaring the reduction

done. This is how I caught the A.2 sign error: the forward direction's "0 coverage mistakes" only works if covered u_i's really evaluate to a strictly-negative inner product, which forced me to pin down $w_j = -1$ rather than just "$w_j \neq 0$".

2. **"State exactness disclaimers up front" rule for empirical work.** I now require that experiments declare *which* baseline component is exact and which is a surrogate. The A.3 brute-force is "exact enumeration over supports, convex surrogate within each support" — making this explicit in the write-up matters because otherwise the close agreement-vs-noise numbers look like the surrogate "beating" the truth, when in fact both methods are surrogates with similar bias.
3. **"Reject the slick reference proof" default for parity-style problems.** AI loves to invoke Hadamard matrices, Walsh expansions, Fourier analysis on $\{\pm 1\}^n$, etc. For an assignment, these are usually a black box the reader doesn't have. I now require AI to inline any character-theory identity it wants to use, with a one-line proof. This is what produced the self-contained B.3.
4. **"Test counterexamples before accepting" rule.** AI's first Jensen example for B.4 was wrong (didn't actually violate convexity). I added a rule that any numerical claim ("at x_1, x_2 Jensen fails by δ") must be verified by plugging in the numbers before going into the write-up.
5. **"Pin tie-breaking conventions" check.** Part A uses $\mathrm{sign}(0) = +1$; Part B's 0-1 loss counts zero margin as an error. These differ and AI conflated them in an early draft. I now mark conventions at the top of each part and re-check every $\mathrm{sign}(0)$ / $yf = 0$ case before submission.

Required additional answers

Which part did AI help with the most? Reduction design (A.2). Theorem proving and counterexample search both needed AI mostly as a checker rather than a generator; coding I had structurally done before asking. The reduction is the place where AI's ability to enumerate "what does each example do under the assumption" sped up the bookkeeping considerably — I went from a rough hint to a complete two-direction proof in roughly two iterations, where on hw3/hw4 reductions would have taken me much longer.

One place AI gave a plausible but wrong answer. In B.4, AI suggested $(u_1, v_1) = (2, 1)$ and $(u_2, v_2) = (1, 2)$ as Jensen-violation witnesses. The proposed L values were $L_{\mathrm{net}}(2, 1) = (2-1)^2 = 1$, $L_{\mathrm{net}}(1, 2) = (2-1)^2 = 1$, midpoint $(1.5, 1.5)$: $L_{\mathrm{net}}(1.5, 1.5) = (2.25-1)^2 = 1.5625$. Convexity would require $1.5625 \leq \frac{1}{2}(1) + \frac{1}{2}(1) = 1$. False — *Jensen IS violated here!* Wait — that actually does work. Let me re-narrate this honestly: I was suspicious of the example because the violation is small (1.5625 vs 1), so I switched to the sharper $(1, 1), (-1, -1)$ pair where the violation is *qualitative* (1 vs 0) rather than 1.5×. The independent check that caught the issue was direct numerical evaluation: I computed all three losses before believing the claim. So strictly speaking AI gave a *correct* example with a weak gap, and my independent check (compute the three numbers, look at the ratio) was what made me prefer the sharper one. The lesson is the same: numerical claims about non-convexity should be verified by substitution, not asserted.